Computerized Medical Imaging and Graphics 110 (2023) 102303

Contents lists available at ScienceDirect

Computerized Medical Imaging and Graphics

journal homepage: www.elsevier.com/locate/compmedimag

# Multimodal transformer network for incomplete image generation and diagnosis of Alzheimer's disease

Xingyu Gao [a], Feng Shi [b,*], Dinggang Shen [b,c,**], Manhua Liu [a,d,***]

[a] School of Electronic Information and Electrical Engineering, Shanghai Jiao Tong University, China
[b] Department of Research and Development, Shanghai United Imaging Intelligence Co.,Ltd., China
[c] School of Biomedical Engineering, ShanghaiTech University, China
[d] MoE Lab of Artificial Intelligence, AI Institute, Shanghai Jiao Tong University, China

A R T I C L E I N F O

*Keywords:*
Multimodal brain images
Generative adversarial network
Transformer
Image generation
Disease diagnosis

A B S T R A C T

Multimodal images such as magnetic resonance imaging (MRI) and positron emission tomography (PET) could provide complementary information about the brain and have been widely investigated for the diagnosis of neurodegenerative disorders such as Alzheimer's disease (AD). However, multimodal brain images are often incomplete in clinical practice. It is still challenging to make use of multimodality for disease diagnosis with missing data. In this paper, we propose a deep learning framework with the multi-level guided generative adversarial network (MLG-GAN) and multimodal transformer (Mul-T) for incomplete image generation and disease classification, respectively. First, MLG-GAN is proposed to generate the missing data, guided by multi-level information from voxels, features, and tasks. In addition to voxel-level supervision and task-level constraint, a feature-level auto-regression branch is proposed to embed the features of target images for an accurate generation. With the complete multimodal images, we propose a Mul-T network for disease diagnosis, which can not only combine the global and local features but also model the latent interactions and correlations from one modality to another with the cross-modal attention mechanism. Comprehensive experiments on three independent datasets (i.e., ADNI-1, ADNI-2, and OASIS-3) show that the proposed method achieves superior performance in the tasks of image generation and disease diagnosis compared to state-of-the-art methods.

## 1. Introduction

Multimodal-based brain images analysis can provide complementary information to enhance the diagnosis of neurodegenerative disorders such as AD and its prodromal stage known as mild cognitive impairment (MCI) (Walhovd et al., 2010; Zhang et al., 2011; Hinrichs et al., 2011; Shi et al., 2018) compared with the single-modal methods (Zhu et al., 2021; Lei et al., 2020; Lian et al., 2022; Fei et al., 2020). Due to the availability of scanners as well as their expensive nature especially in PET, multimodal brain images are often incomplete in clinical practice. Existing multimodal analysis methods often work in three ways: (1) only a subset of samples with complete multimodal images are studied (Dai et al., 2012; Suk et al., 2014; Hao et al., 2020; Cheng et al., 2019; Zhou et al., 2019b); (2) according to the availability of modalities, samples are divided into several groups, and each group holds a diagnosis model (Xiang et al., 2014; Thung et al., 2014; Liu et al., 2017; Liu et al., 2018b; Zhou et al., 2020; Han et al., 2022); (3) missing data are estimated with the available modality, thus all samples could be used in the analysis (Nie et al., 2016; Huynh et al., 2016; Ben-Cohen et al., 2019; Yang et al., 2020; Hu et al., 2020; Dar et al., 2019; Pan et al., 2019; Gao et al., 2022).

The most straightforward method for multimodal analysis is to discard the samples with missing data. Dai et al. (2012) proposed a multi-classifier framework to combine multimodal features of structural and functional MRI for AD diagnosis. Due to the limited computation, a deep learning method based on the Boltzmann machine was devised for hierarchical latent space representation on paired MRI and PET patches (Suk et al., 2014). In terms of consistent constraint modeling, Hao et al. (2020) proposed a multimodal feature selection method for AD and MCI analysis, which extracted the brain ROIs as the input features rather than patches of MRI and PET. Besides that, a multi-label transfer learning

* Corresponding author.
** Corresponding author at: Department of Research and Development, Shanghai United Imaging Intelligence Co.,Ltd., China.
*** Corresponding author at: School of Electronic Information and Electrical Engineering, Shanghai Jiao Tong University, China.
*E-mail addresses:* feng.shi@uii-ai.com (F. Shi), Dinggang.Shen@gmail.com (D. Shen), mhliu@sjtu.edu.cn (M. Liu).

https://doi.org/10.1016/j.compmedimag.2023.102303
Received 16 July 2022; Received in revised form 27 June 2023; Accepted 27 September 2023
Available online 30 September 2023
0895-6111/

model was presented to extract the high-level features from MRI and cerebrospinal fluid (CSF) data, then a SVM classifier was used for disease classification (Cheng et al., 2019). And Zhou et al. (2019b) proposed a novel deep neural network for dementia diagnosis, which consists of three stages with each stage learning the feature representations from different combinations of MRI, PET, and single nucleotide polymorphism (SNP). The above studies only used the modality-complete samples, which significantly reduced the number of studied subjects.

Another category of methods tries to use all samples including those with missing modalities by dividing the incomplete data into several groups according to availability. Xiang et al. (2014) proposed a bi-level learning method for simultaneously performing the feature-level and source-level analysis of incomplete data from ADNI. This method first transformed multimodal brain images into a matrix consisting of model scores and missing entries. Then, the missing entries were estimated by available modalities. Finally, the complete score matrix was used to predict the clinical level of patients. Similarly, a matrix shrinkage and completion method was proposed for AD diagnosis (Thung et al., 2014). The main contribution is to estimate a high-level matrix of prediction scores rather than the missing low-level original images. Liu et al. (2017) proposed a view-aligned hypergraph learning method for AD diagnosis. In this method, three original data (i.e., MRI, PET, and CSF) were firstly divided into several views according to availability. Then, view-aligned hypergraph classifiers (VAHC) were constructed in each view space based on sparse representation. With the generated class probability scores by VAHC, a multi-view label fusion method was devised to make the final classification. They further constructed another hypergraph model via the strategy of star expansion, which replaced the VAHC with hypergraph Laplacian computation for AD/MCI prediction (Liu et al., 2018b). Besides that, Zhou et al. (2020) proposed a multimodal latent space inducing network for early AD diagnosis with MRI and PET images, which consists of two models (i.e., CMLS and IMLS) for dealing with the complete and incomplete multi-modality data, respectively. The above methods can make use of all available data by modeling the underlying high-order relationships between different modalities. However, these methods need to build multiple models for the groups with different available modalities, which degrades the computation efficiency especially when the number of modalities is large. In addition, for the incomplete samples, there was no extra information to be introduced for the missing data, which may affect the diagnostic efficacy.

For the challenges of modality incomplete, a promising alternative way is to estimate the missing data via machine learning methods. Nie et al. (2016) proposed a 3D fully convolutional network (FCN) to learn an end-to-end nonlinear mapping for estimating CT images from MRI, which could better preserve the neighborhood information of generated CT images than the conventional convolutional neural network (CNN). Another method is to predict CT patches with the MRI (Huynh et al., 2016). This method first applied a patch-wise structured random forest for the initial prediction of CT patches and then constructed an auto-context model to iteratively refine the prediction. Finally, all of the predicted CT patches were assembled to obtain the final whole prediction.

Recently, generative adversarial networks (GAN) become a successful paradigm for estimating the missing data for multimodal analysis. Ben-Cohen et al. (2019) presented a FCN-based GAN to generate PET images from the given CT scans. The generated PET images were further used to reduce the false-positive rate in lesion detection. Besides that, a structure-constrained Cycle-GAN was designed for MRI-to-CT synthesis through unsupervised learning (Yang et al., 2020). In this method, a modality-independent neighborhood descriptor (MIND) was proposed to compute the non-local patch-based self-similarity for the structural features. Furthermore, an extra structure- consistency loss was defined to encourage the two sets of voxel-wise MIND features to be similar. Results on unpaired abdomen MRI-to-CT generation show promising performance.

For the estimation of missing brain images, Hu et al. (2020) proposed a bidirectional GAN for the synthesis of PET with MRI. This bidirectional mapping mechanism embedded the diverse structural details into a high-level latent space. In the training process, additional pixel-wise loss and perceptual loss were introduced to yield more realistic results. To better learn the nonlinear intensity transformation between the source and target images, a novel conditional GAN was proposed for multi-contrast MRI synthesis (Dar et al., 2019), which consists of two stages: First, the network was trained with the pixel-wise loss and perceptual loss between the generated and ground truth images; Then, the pixel-wise loss was replaced with a cycle loss to enhance the ability of target images reconstruction. To pay more attention to the disease-specific information conveyed in multimodal brain images, Pan et al. (2019) proposed a disease-specific GAN for AD/MCI diagnosis. They first built an independent disease-specific neural network (DSNN) for disease classification, and then this DSNN was used to aid the image generation with Cycle-GAN. This method improved the diagnostic efficacy of the generated brain images. Although many efforts were devoted to addressing the problem of data incomplete, it is still hard to make use of the missing data and multimodal correlations for disease diagnosis. To solve this problem, a task-induced pyramid and attention GAN was designed to generate the missing PET with its corresponding MRI (Gao et al., 2022). The pyramid convolution layers and attention module as well as the disease classification task were combined into the basic GAN. Then, a dense convolution network with path-wise transfer blocks was proposed for the final disease classification.

In this paper, we propose a framework with multi-level guided GAN (MLG- GAN) and multimodal transformer (Mul-T) for brain image generation and classification, respectively. Towards a more accurate generation for disease diagnosis, MLG-GAN was proposed to estimate the missing data, guided by multi-level disease-related information from voxels, features, and tasks. With the complete multimodal images, a Mul-T is developed to model the latent interactions and correlations from one modality to another with the cross-modal attention mechanism for disease diagnosis. Comprehensively, our contributions can be summarized as follows:

- In the generation of the missing data, we first propose a feature auto-regression branch to embed the high-level features of target images into the generator of GAN for a more accurate image generation. Second, we build a novel task-level discriminator with transformer blocks to model the spatial relationships of high-level features. Finally, in addition to the global supervision, we include voxel-level supervision on the local salient regions for the generation of local abnormal regions caused by disease.
- In multimodal disease diagnosis, we propose a novel multimodal transformer network to combine the multi-level and multimodal features. First, in addition to the global branch for the extraction of global features, a local network branch with spatial attention is devised. Then, a multimodal transformer with cross self-attention is investigated for disease classification.

The rest of this paper is organized as follows. In Section 2, we describe the dataset used in this study and introduce the data preprocessing steps. Subsequently, we give a detailed description of our proposed method in Section 3. In Section 4, we show the experimental results. Discussion and conclusion are presented in Section 5 and Section 6, respectively.

## 2. Materials

### 2.1. Dataset

The multimodal brain images used in this study are obtained from two public databases: the Alzheimer's Disease Neuroimaging Initiative (ADNI) (Jack et al., 2008) and the Open Access Series of Imaging Studies

**Table 1**
Demographic characteristics and clinical information of the studied subjects from ADNI-1/2 and OASIS-3.

| Dataset | Category | Gender (Female/ Male) | Total Number w/ Complete Data | Age (Years) | CDR | MMSE |
|---|---|---|---|---|---|---|
| ADNI-1 | AD | 97/99 | 92 | 75.7 ± 7.7 | 0.8 ± 0.3 | 23.3 ± 2.0 |
| | pMCI | 66/102 | 76 | 74.8 ± 6.9 | 0.5 ± 0.0 | 26.6 ± 1.7 |
| | sMCI | 77/153 | 127 | 75.0 ± 7.7 | 0.5 ± 0.0 | 27.3 ± 1.8 |
| | NC | 110/117 | 100 | 75.9 ± 5.0 | 0.0 ± 0.0 | 29.1 ± 1.0 |
| ADNI-2 | AD | 66/90 | 144 | 75.7 ± 7.9 | 0.9 ± 0.4 | 21.8 ± 3.8 |
| | pMCI | 31/35 | 65 | 72.9 ± 7.3 | 0.6 ± 0.2 | 25.7 ± 2.2 |
| | sMCI | 50/62 | 93 | 72.6 ± 7.9 | 0.5 ± 0.1 | 27.7 ± 2.2 |
| | NC | 106/94 | 185 | 74.1 ± 6.3 | 0.0 ± 0.0 | 28.9 ± 1.3 |
| OASIS-3 | AD | 85/89 | 126 | 75.2 ± 8.9 | 0.7 ± 0.3 | 24.6 ± 4.2 |
| | NC | 69/102 | 123 | 72.6 ± 8.0 | 0.0 ± 0.0 | 29.0 ± 1.3 |

(OASIS) (LaMontagne et al., 2019). For the ADNI database, its goal is to develop the relations between biomarkers and clinical trials for AD diagnosis. Two subsets are used in this study (i.e., ADNI-1 and ADNI-2). For the OASIS database, its goal is to facilitate future discoveries in basic and clinical neuroscience. The OASIS-3 subset consists of multimodal longitudinal neuroimaging and clinical information on CN and AD. In this study, the baseline T1w-MRI, T2w-MRI, and FDG-PET images are collected to train and evaluate our method. There are 821 subjects in ADNI-1, including 196 AD, 227 CN, 168 pMCI, and 230 sMCI. In ADNI-2, there are 534 subjects, including 156 AD, 200 CN, 66 pMCI, and 112 sMCI. pMCI and sMCI are categorized based on the clinical follow-up data. In particular, patients with pMCI are defined as initially diagnosed with MCI and transited to AD within a span of 36 months. While patients with sMCI are defined as initially diagnosed with MCI but not progressed to AD within the same timeframe. In OASIS-3, there are 345 subjects, including 174 AD, and 171 CN. All subjects in ADNI-1/2 and OASIS-3 have T1w-MRI, while only part of them have T2w-MRI and FDG- PET. The demographic and clinic information of participants from ADNI-1/2 and OASIS-3 are shown in Table 1. More details about these databases could be found on official websites.

### 2.2. Image pre-processing

The original T1w-MRI, T2w-MRI, and FDG-PET images are pre-processed by FSL, which can be downloaded from the website: https://fsl.fmrib.ox. ac.uk/. First, BET algorithm is used to strip the skull and extract the whole brain region (Smith, 2002), Briefly, the intensity histogram is first processed to determine a robust brain/non-brain threshold. The head centroid is then determined according to the rough size of 3D head image. A triangular mesh of the spherical surface is initialized inside the brain and allowed to slowly change its shape. The reshaping follows forces that maintain the surface space and smoothness, while simultaneously trying to extend towards the edge of the brain. Finally, the outer surface of the brain ROI is estimated and used for the skull-stripping of the input image. Second, FLIRT linear registration algorithm is used to align the stripped images to the MNI standard template space with the size of 182 × 182 × 182 (Jenkinson et al., 2002), from which linear interpolation is used to reconstruct a full continuous image from the discrete points. In terms of inter-model voxel similarity registration, a global and local optimization method is used for cost function optimization. Finally, the initial skull-stripped brain image is transformed into the MNI standard template space. To alleviate the computation and memory costs, we further remove the zero values around the whole brain regions and reduce the size of all T1w-MRI, T2w-MRI, and PET images into 76 × 94 × 76 as the inputs of our

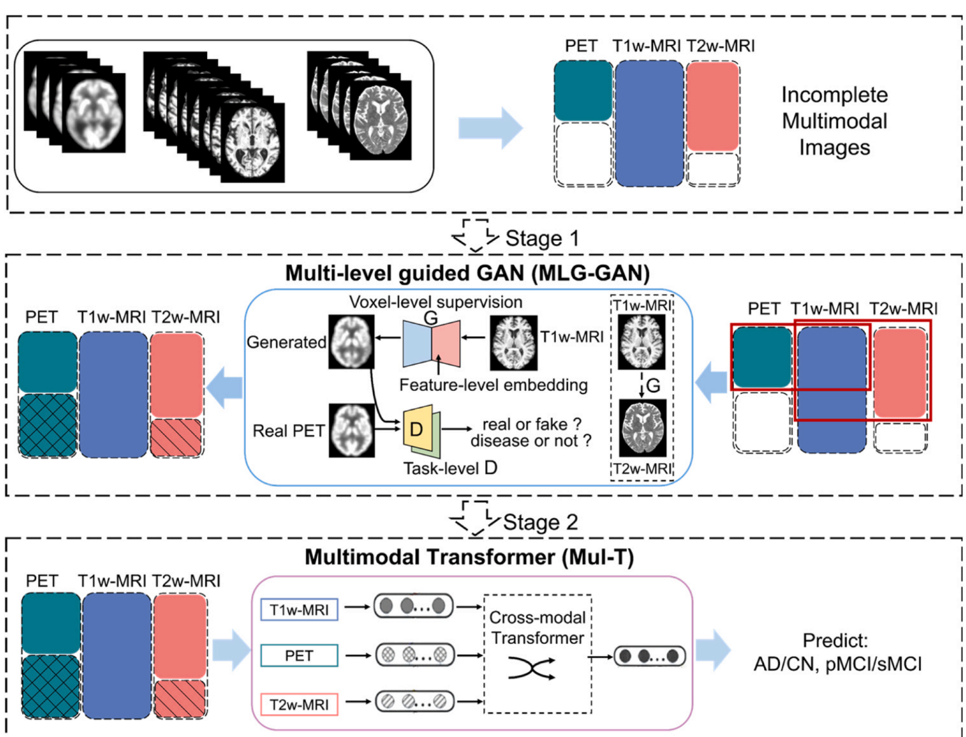


**Fig. 1.** Overview of proposed deep learning framework with multi-level guided GAN (MLG- GAN) and multimodal transformer (Mul-T) for brain image generation and classification, respectively.

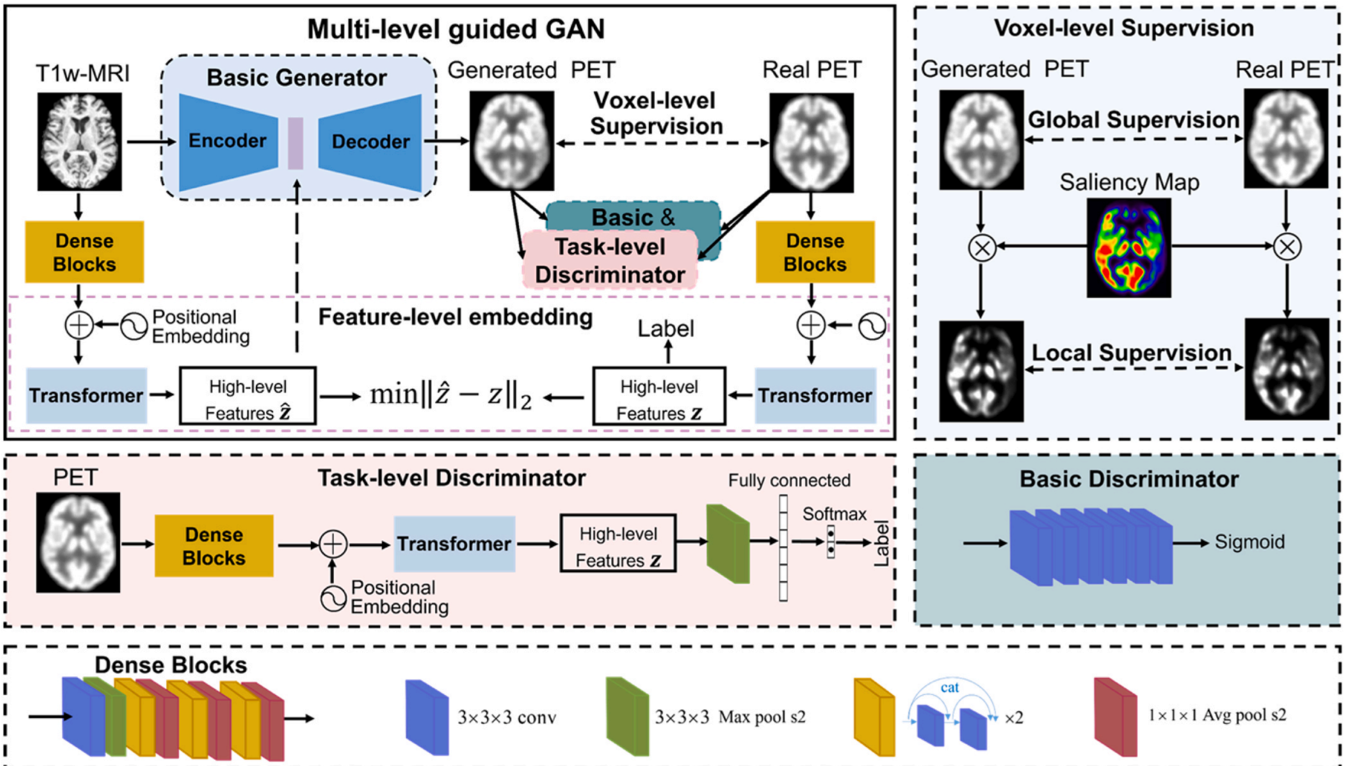


**Fig. 2.** Diagram of our proposed multi-level guided GAN (MLG-GAN) for brain imaging generation. The generations of PET and T2w-MRI with T1w-MRI share the same architecture.

method.

## 3. Methods

### 3.1. Overview of the proposed method

It is well known that incomplete data and multimodal fusion are the major challenges of multimodal image analysis. To address these problems, we propose a deep learning framework that consists of a multi-level guided GAN (MLG-GAN) and a multimodal transformer (Mul-T) for the generation of missing images and the classification of multimodal features, respectively. Fig. 1. illustrates the flowchart of our proposed method. The details of the network architectures are presented in the following subsections.

### 3.2. Multi-level guided GAN (MLG-GAN) for image generation

Generative adversarial network (GAN), which consists of generator and discriminator, has been widely investigated for image generation. The generator aims to generate the real ones as closely as possible, while the discriminator tries to distinguish the generated image from its corresponding ground truth. This process is iteratively repeated until the generated image is close enough to the real ones and thus the discriminator can no longer distinguish them.

In our study, the most challenging work is to generate the missing data with considerable sensitivity for the diagnosis task. So that the generated images could provide disease-related information for diagnosis. Since the basic GAN can only conform to the pixel distribution and visual quality, it ignores the reconstruction of abnormal changes caused by disease. More importantly, brain disease may cause abnormalities in different regions from different imaging modalities. It is hard to estimate the disease-related information due to the limited data on the disease. To solve these problems, we propose a multi-level guided GAN (MLG-GAN) for multimodal brain image generation, which can make full use of the multi-level disease-related information from the components of task-level discriminator, feature-level embedding, and voxel-level supervision. The architecture of MLG-GAN is illustrated in Fig. 2, and the details of each component are depicted as follows.

#### 3.2.1. Basic GAN

Generally, the basic GAN consists of a generator and discriminator. The U-Net structure (Gao et al., 2022) is adopted in our generator. The encoding part aims to extract vital features from the image of the original domain. It contains 4 blocks, and each block consists of two $3 \times 3 \times 3$ convolution layers followed by a rectified linear unit (ReLU). One $2 \times 2 \times 2$ max-pooling layer is used for feature down-sampling between two blocks. Finally, 6 residual blocks are added to the last block. The decoding part takes the features learned from the encoding part to reconstruct the image of the target domain. Its architecture is similar to the encoding part, the difference is that the down-sampling layers with max-pooling are replaced by the up-sampling layers with de-convolution.

The discriminator of basic GAN is built with a lightweight CNN to distinguish the generated image from the real one according to the voxel distribution and visual quality. It consists of 6 convolution layers to extract features and make discrimination.

Let $X_o^i$, $X_t^i$be the brain images of $i^{th}$ subject in the original domain and target domain, respectively. Since the T1w-MRI are easy to capture, the brain image in the source domain is T1w-MRI while the target domain images are PET and T2w-MRI. The loss function of basic GAN is $L_{GAN} = L_G + L_D$, where $L_G$ and $L_D$are the generation loss and discrimination loss, respectively, formulated as follows:

$$L_G = E_{x\sim P_o(x)}\left(\log\left(I - D\left(G\left(X_o^i\right)\right)\right)\right) \tag{1}$$

$$L_G = E_{x\sim P_t(x)}\left[\log\left(I - D\left(X_t^i\right)\right)\right] + E_{x\sim P_o(x)}\left[\log\left(D\left(G\left(X_o^i\right)\right)\right)\right] \tag{2}$$

where $G$denotes the generator, and $D$ denotes the discriminator.

#### 3.2.2. Task-level discriminator

The basic GAN considers the voxels of the whole brain equally that can estimate extensive and general contents. However, it ignores the subsequent disease classification task. Thus, the basic GAN cannot work well to reconstruct the details of abnormal information about the characteristics of the disease. In this work, we propose a task-level discriminator for the basic GAN, which consists of dense blocks followed by a transformer block, as shown in Fig. 2. The transformer block employs the self-attention mechanism to capture the long-range spatial correlations of high-level features. Then the high-level features are condensed by a 3 × 3 × 3 max-pooling layer with stride as 2. Finally, a fully connected layer with SoftMax activation is used to output the predicted label. This component is devised to force the generated images to hold the task-relevant information that makes the subsequent multi-modal classification model can discriminate the disease subjects from normal ones easily. The cross-entropy loss is used for the task-level discriminator and computed as:

$$\boldsymbol{L}_{Dtask} = \boldsymbol{E}_{x\sim P_o(x)}\left[-\boldsymbol{y}_i * \log\left(\boldsymbol{D}_{task}\left(\boldsymbol{G}\left(\boldsymbol{X}_o^i\right)\right)\right) - (\boldsymbol{1}-\boldsymbol{y}_i) * \log\left(\boldsymbol{1} - \boldsymbol{D}_{task}\left(\boldsymbol{G}\left(\boldsymbol{X}_o^i\right)\right)\right)\right] \tag{3}$$

where $\boldsymbol{D}_{task}$ denotes the task-level discriminator, which outputs the predicted label; $\boldsymbol{y}_i$ denotes the real label of the $\boldsymbol{i}^{th}$ subject. The task-level discriminator provides feedback from the disease diagnosis for image generation.

#### 3.2.3. Feature-level embedding

In addition to task-level supervision, the feature-level constraint can also provide an improvement for image generation. Pan et al. (2019) designed a feature-consistent constraint on the multi-level features from the synthetic and real images. Although the constraints can ensure an accurate estimation, the generation process is still a one-direction prediction, i.e., predicting each voxel value in the target domain with original images. Different from that, we embed additional high-level features learned from the target domain images into the generator for image generation. Specifically, we introduce a feature auto-regression branch to predict the features of the target domain through learning knowledge from the original domain. The feature auto-regression branch consists of dense convolution blocks and transformer blocks, as shown in Fig. 2. To capture more disease-related features from the target domain, we first pre-train the right-side regression branch for disease diagnosis and then fine-tune the whole network for features regression, which can transform the high-level features from the source domain to the target domain. Then, the predicted features are further embedded into the generator for more accurate image generation. The feature-level regression loss is calculated as:

$$\boldsymbol{L}_{Feat} = \| \hat{z} - z \|_2 \tag{4}$$

where $\hat{z}$ and $z$ denote the generated high-level features and the ground truths, respectively.

#### 3.2.4. Voxel-level supervision

In addition to the feature-level embedding, we apply voxel-level supervision to encourage the consistency of voxel values between the generated images and the ground truths. The voxel-level supervision includes two parts: global supervision treats all voxels of the whole brain equally, while local supervision focuses on the salient regions associated with the disease. The salient regions are obtained through the saliency map which is calculated by the task-level discriminator. For better supervision, we combine the $\boldsymbol{L_1}$ and $\boldsymbol{SSIM}$ loss to the final loss function of voxel-level supervision, which can be computed as:

$$\boldsymbol{L}_{Vox} = \boldsymbol{M}[\ \| \hat{\boldsymbol{X}} - \boldsymbol{X} \|_{\boldsymbol{1}} + \boldsymbol{SSIM}(\hat{\boldsymbol{X}}, \boldsymbol{X})] + \ \| \hat{\boldsymbol{X}} - \boldsymbol{X} \|_{\boldsymbol{1}} + \boldsymbol{SSIM}(\hat{\boldsymbol{X}}, \boldsymbol{X}) \tag{5}$$

where $\hat{\boldsymbol{X}}$ and $\boldsymbol{X}$ denote the generated image and the ground truth, respectively. $\|\cdot\|_t$ is the norm operator; the $\boldsymbol{SSIM}$ function is calculated as those in Wang et al. (2004); $\boldsymbol{M}$ represents the normalized weight of the saliency map, which can be expressed as:

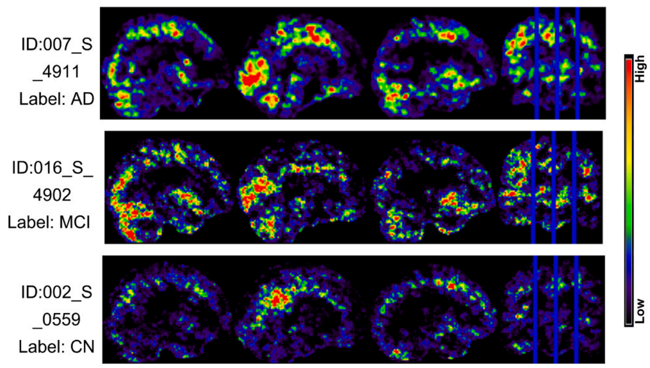


**Fig. 3.** Saliency maps across AD, MCI, and CN subjects on the PET images corresponding to Fig. 2.

$$\boldsymbol{M} = \left\langle \frac{\partial \boldsymbol{S}_c(\boldsymbol{X}_i)}{\partial \boldsymbol{X}_i} \right\rangle \tag{6}$$

where $\langle\cdot\rangle$ denotes the min-max normalization operation, for making the value range of $\boldsymbol{M}$ to [0,1]; $\boldsymbol{S_c(X_i)}$ is the score function of the classification model for input image $\boldsymbol{X_i}$ assigned to class $\boldsymbol{c}$, can be calculated as:

$$\boldsymbol{S}_c(\boldsymbol{X}_i) = \boldsymbol{W}^T\boldsymbol{X}_i + \boldsymbol{b}_c \tag{7}$$

where $\boldsymbol{W_c}$ and $\boldsymbol{b_c}$ are the weight vector and bias of the classification model, respectively. In our work, the generator will first focus on the local regions associated with the disease, and then pay more attention to global supervision. Fig. 3 presents three saliency maps of AD, MCI, and CN subjects on the PET images corresponding to Fig. 2.

#### 3.2.5. Total loss

To combine the multi-level guided components with respect to the task-level discriminator, feature-level embedding, and voxel-level supervision, we define the total loss of our MLG-GAN as:

$$\boldsymbol{L} = \boldsymbol{\lambda L}_{GAN} + \boldsymbol{\eta L}_{Dtask} + \boldsymbol{\beta L}_{Feat} + \boldsymbol{\sigma L}_{Vox} \tag{8}$$

where $\lambda$, $\boldsymbol{\eta}$, $\boldsymbol{\beta}$ and $\boldsymbol{\sigma}$ are the coefficients of each individual loss. The coefficient of $\boldsymbol{L_{GAN}}$ loss can be automatically calculated as:

$$\boldsymbol{\lambda} = 10^{\left\lfloor \log \frac{Max\{L_{GAN}, L_{Dtask}, L_{Feat}, L_{Vox}\}}{L_{GAN}} \right\rfloor} \tag{9}$$

where $\lfloor\cdot\rfloor$ denotes the floor operation. Accordingly, $\boldsymbol{\eta}$, $\boldsymbol{\beta}$ and $\boldsymbol{\sigma}$ are calculated in the same way. The weights of hyperparameters are important for the training process, for instance, if $\boldsymbol{\lambda L_{GAN}}$ is far greater than $\boldsymbol{\eta L_{Dtask}}$, the training process might pay more attention to the voxel-level image generation, potentially ignore the status (i.e., disease or normal) of the generated output. On the contrary, the generated image could obtain accurate status but with poor voxel-level image quality. In summary, this strategy is set to adjust the losses to the same magnitude adaptively, which can balance the training process of generation. To further speed up the training process, we first pre-train the task-level discriminator and feature-level embedding, independently. Then we train the generator and discriminator of Basic GAN with the voxel supervision iteratively.

### 3.3. Multimodal transformer (Mul-T) for disease classification

After the missing PET and T2w-MRI are generated by our MLG-GAN, the next step is to extract and combine the features from complete multimodal brain images (i.e., T1w-MRI, PET, and T2w-MRI) for disease

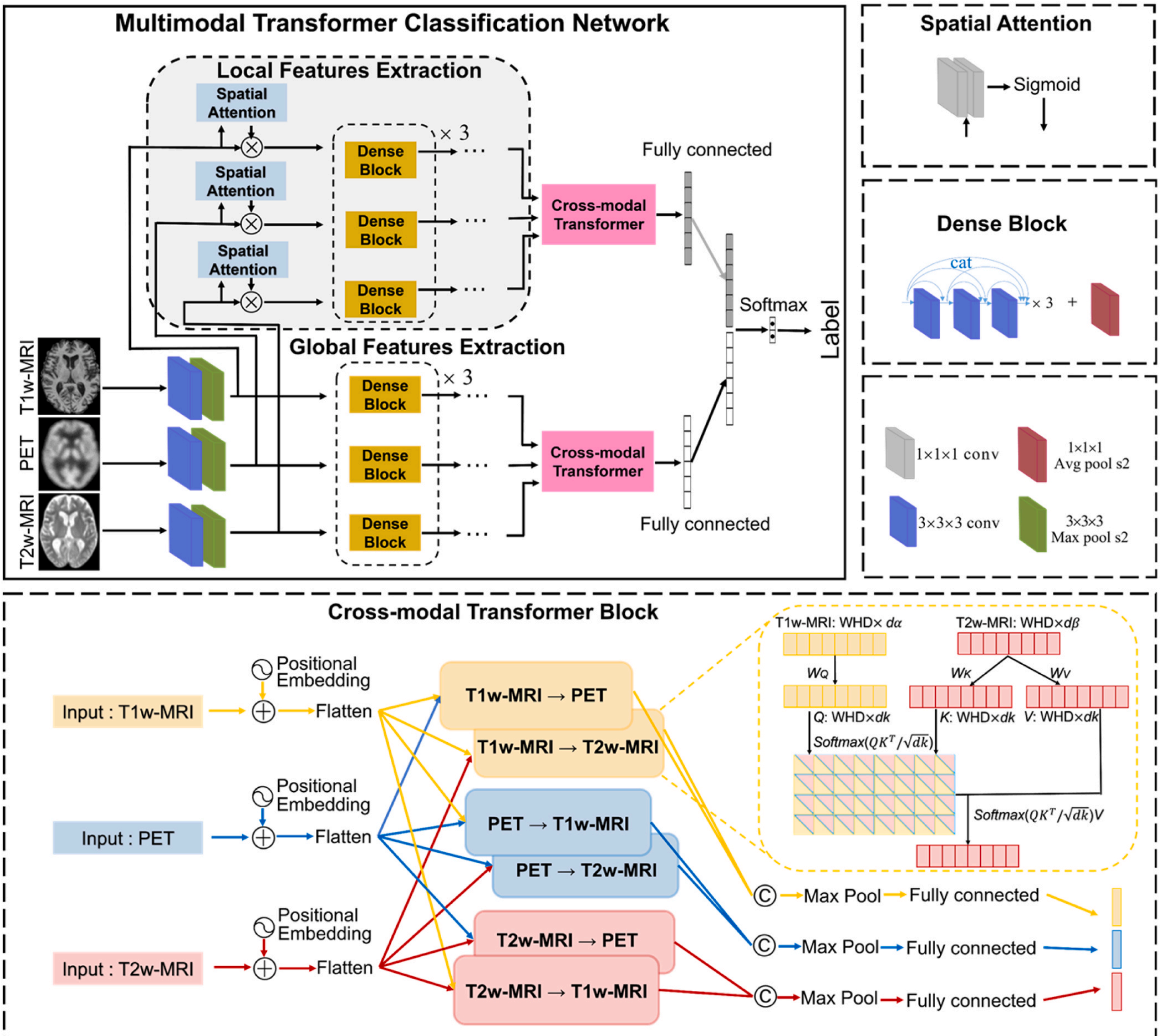


**Fig. 4.** Architecture of our proposed Multimodal Transformer (Mul-T) for classifications of AD/CN or pMCI/sMCI.

diagnosis. To this end, we propose a multimodal transformer (Mul-T) network, as shown in Fig. 4. The network consists of two main parts: global and local feature extractions, cross-modal transformer, which will be presented in detail as follows.

### 3.3.1. Global and local feature extraction

To extract the features of multimodal brain images, we build a deep convolutional neural network (CNN) consisting of one convolution layer followed by a $3 \times 3 \times 3$ max-pooling layer and 3 dense blocks for each modality. The dense blocks are used to extract the disease-related information from low-level source modality to high-level latent space. The architecture of deep CNN is shown in Fig. 4. To extract rich features, we build the global and local branches on brain images. The global branches aim to learn features from the whole brain while the local branches try to learn features from the local regions which are obtained through the spatial attention operation. Since the global branches may lose some important information related to the disease, the spatial attention module is embedded at the low-level stage to hold the vital information. To ensure effectiveness, all CNN branches are individually pre-trained on each modality (i.e., T1w-MRI, PET, or T2w-MRI).

### 3.3.2. Cross-modal transformer

After extracting the global and local features for each modality, the next step is to combine the multimodal information for disease diagnosis. Different from the simple concatenation at the fully connected stage in most existing methods, we propose a cross-modal transformer block on the global and local branches to model the interactions and correlations between different modalities, which is inspired by Tsai et al. (2019). The architecture of the cross-modal transformer block is shown at the bottom of Fig. 4. In each cross-modal transformer, this cross-modal attention can model the latent interactions and adaptions from one modality to another. Let $\boldsymbol{H}_\alpha$ and $\boldsymbol{H}_\beta$ be the feature vectors of two modalities $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$, respectively. With the cross-modal attention mechanism, **Queries** are from the modality $\boldsymbol{\alpha}$ and defined as $\boldsymbol{Q} = \boldsymbol{W}_Q \boldsymbol{H}_\alpha$, **Keys** and **Values** are from another modality $\boldsymbol{\beta}$ and defined as $\boldsymbol{K} = \boldsymbol{W}_K \boldsymbol{H}_\beta$ and $\boldsymbol{V} = \boldsymbol{W}_V \boldsymbol{H}_\beta$, respectively, where $\boldsymbol{W}_Q$, $\boldsymbol{W}_K$, $\boldsymbol{W}_V$ are the projection matrices. Then the output of cross-modal attention can be computed as:

$$C = SoftMax\left(QK^{T} \Big/ \sqrt{d_k}\right)V \tag{10}$$

Since the transformer works in parallel, it requires additional positional information to learn the correlation among image voxels. To enable the features to carry positional information, positional embedding (***PE***) is attached to the inputs prior to the flattening operation. The ***PE*** can be computed as:

$$PE(pos, 2i) = sin\left(pos/10000^{2i/d}\right) \tag{11}$$

$$PE(pos, 2i+1) = \cos\left(pos/10000^{2i/d}\right) \tag{12}$$

here, ***pos*** and ***i*** represent the row and column of the input feature, respectively.

***d*** denotes the overall size of the feature vector. Finally, there are 6 cross-modal transformer subsets on 3 modalities in total. Through a $3 \times 3 \times 3$ max-pooling layer and a fully connected layer, we concatenate the outputs and then apply a SoftMax activation layer for the final disease classification.

Our Mul-T has two significant benefits: (1) Global and local feature extraction could contain comprehensive information related to the disease, and thus provides ideal inputs for subsequent cross-model transformer; (2) Cross-model transformer can handle long-term dependencies in feature sequences and does not suffer from vanishing or exploding gradients as recurrent neural networks (RNNs) often do. Furthermore, the self-attention mechanism in Mul-T allows the model to weigh the importance of each element from cross-modalities. This makes the model interpretable and robust.

## 4. Experiments

In this section, the experiments will be conducted to evaluate the proposed method for the generation of missing data and the classification of neurodegenerative disease.

### 4.1. Experimental settings

Our proposed algorithm is implemented with Python 3.7.4 on the Pytorch framework. The calculating platform is Ubuntu 16.04 with a CPU of Intel Xeon E5–2620 and a GPU of NVIDIA GeForce RTX 2080 Ti. The proposed method consists of MLG-GAN and Mul-T. To better evaluate the generalization, we train the proposed models on ADNI-1 and test them on two independent datasets: ADNI-2 and OASIS-3. Specifically, in image generation by MLG-GAN, we use all available T1w-MRI, PET, and T2w-MRI for training and testing. For the multimodal disease classification with Mul-T, the missing modalities are generated and included for training and testing.

To build the MLG-GAN, we first pre-train the task-level discriminator on the real target images. Then, we train the feature-level auto-regression branch to predict the high-level features of target images. Finally, the generation network is trained with the pre-trained task-level discriminator and feature-level auto-regression branch to optimize the weights of the generator. Adam solver is adopted in the training stage with a batch size of 4 in both subnetworks. The learning rates of the task-level discriminator and feature-level auto-regression branches are both set to 0.001. For the basic generator and discriminator, the learning rates are set to 0.0001 and 0.0004, respectively.

To build the Mul-T, we first pre-train the global and local CNN branches for feature extraction, then we introduce the cross-modal transformers to fine- tune the whole classification network. Adam solver is used as the optimizer, batch size and learning rate are set to 4 and 0.001, respectively. We test our Mul-T on classifications of AD vs NC and pMCI vs sMCI. These two tasks are highly relevant, and the classification of AD vs NC is easier than pMCI vs sMCI. To facilitate the training process, we first train our Mul-T on AD vs NC, and then transfer the learned parameters to initialize the model of pMCI vs sMCI classification.

**Table 2**
Comparison with state-of-the-art methods and ablation studies for the generated PET images on ADNI-2, where the models are trained on ADNI-1.

| Method | SSIM | PSNR | MSE | MMD | AUC* | AUC⁺ |
|---|---|---|---|---|---|---|
| Cycle-GAN (Pan et al., 2018) | 0.880 | 27.2 | 388 | 0.249 | 68.0 | 62.7 |
| TPA-GAN (Gao et al., 2022) | 0.915 | 29.0 | 184 | 0.107 | 88.3 | 72.4 |
| Basic GAN | 0.868 | 26.1 | 392 | 0.224 | 61.4 | 55.6 |
| Basic GAN + V | 0.920 | 28.3 | 174 | 0.104 | 79.8 | 64.3 |
| Basic GAN + V + FE | 0.920 | 29.1 | 168 | 0.086 | 89.0 | 73.2 |
| Our MLG-GAN | **0.923** | **29.1** | **165** | **0.083** | **92.4** | **75.1** |

**Table 3**
Comparison with state-of-the-art methods and ablation studies for the generated T2w-MRI images on OASIS-3, where the models are trained on ADNI-1.

| Method | SSIM | PSNR | MSE | MMD | AUC* |
|---|---|---|---|---|---|
| Cycle-GAN (Pan et al., 2018) | 0.756 | 26.6 | 413 | 0.314 | 86.1 |
| TPA-GAN (Gao et al., 2022) | 0.799 | **27.1** | 360 | 0.216 | 87.0 |
| Basic GAN | 0.803 | 25.3 | 381 | 0.457 | 81.6 |
| Basic GAN + V | 0.813 | 26.0 | 377 | 0.222 | 84.3 |
| Basic GAN + V + FE | 0.815 | 26.1 | 368 | 0.178 | 87.5 |
| Our MLG-GAN | **0.819** | 26.3 | **357** | **0.167** | **89.4** |

### 4.2. Results of image generation

The first experiment is to test the effectiveness of image generation, we compare our method with other state-of-the-art methods, including the Cycle- consistent GAN (Cycle-GAN) (Pan et al., 2018) and Task-induced Pyramid and Attention GAN (TPA-GAN) (Gao et al., 2022). For a fair comparison, these methods are trained and tested on the same datasets. Table 2 illustrates the average results of all testing subjects from ADNI-2 dataset for T1w-MRI to PET generation. Higher values of structural similarity index measure (***SSIM***) and peak signal-to-noise ratio (***PSNR***) indicate better image quality, and the measures of mean squared error (***MSE***) and maximum mean discrepancy (***MMD***) are on the contrary. The area under the receiver operating characteristic (***AUC***) is used to test the classification performances of generated images of AD/CN (***AUC****) and pMCI/sMCI ($AUC^{+}$) by task-level discriminator, which has already been pre-trained on real images. Accordingly, Table 3 illustrates the results of T1w-MRI to T2w-MRI generation for the testing subjects from OASIS-3. From the results, we can see that our method achieves the best performance on both T1w-MRI to PET and T1w-MRI to T2w-MRI generations.

Besides that, we conduct a series of ablation studies including the U-Net based “Basic GAN”, and its improvements by gradually adding the voxel-level supervision, feature-level embedding, and the task-level discriminator, which are denoted as “Basic GAN + V”, “Basic GAN + V + FE” and “Our MLG- GAN”, respectively. The results on ADNI-2 and OASIS-3 datasets (shown in Table 2 and Table 3, respectively) show that the generated PET and T2w-MRI are gradually improved by introducing the voxel-level supervision, feature-level embedding, and task-level discriminator into the “Basic GAN”. Especially, the ***AUC*** indicators are greatly improved. That suggests more discriminative information about the disease can be generated by our method. Furthermore, we compare the generated PET and T2w-MRI through our MLG-GAN with its ablation studies as well as other state-of-the-art methods for two subjects from the testing sets, as illustrated in Fig. 5. From the results, we can see that the generated images with the Basic GAN have poor image quality due to the larger area of high residuals. Benefitting from the voxel-level supervision, the “Basic GAN + V” can improve the quality of generated images. The feature- level embedding can produce a great improvement, while the TPA-GAN also generates a

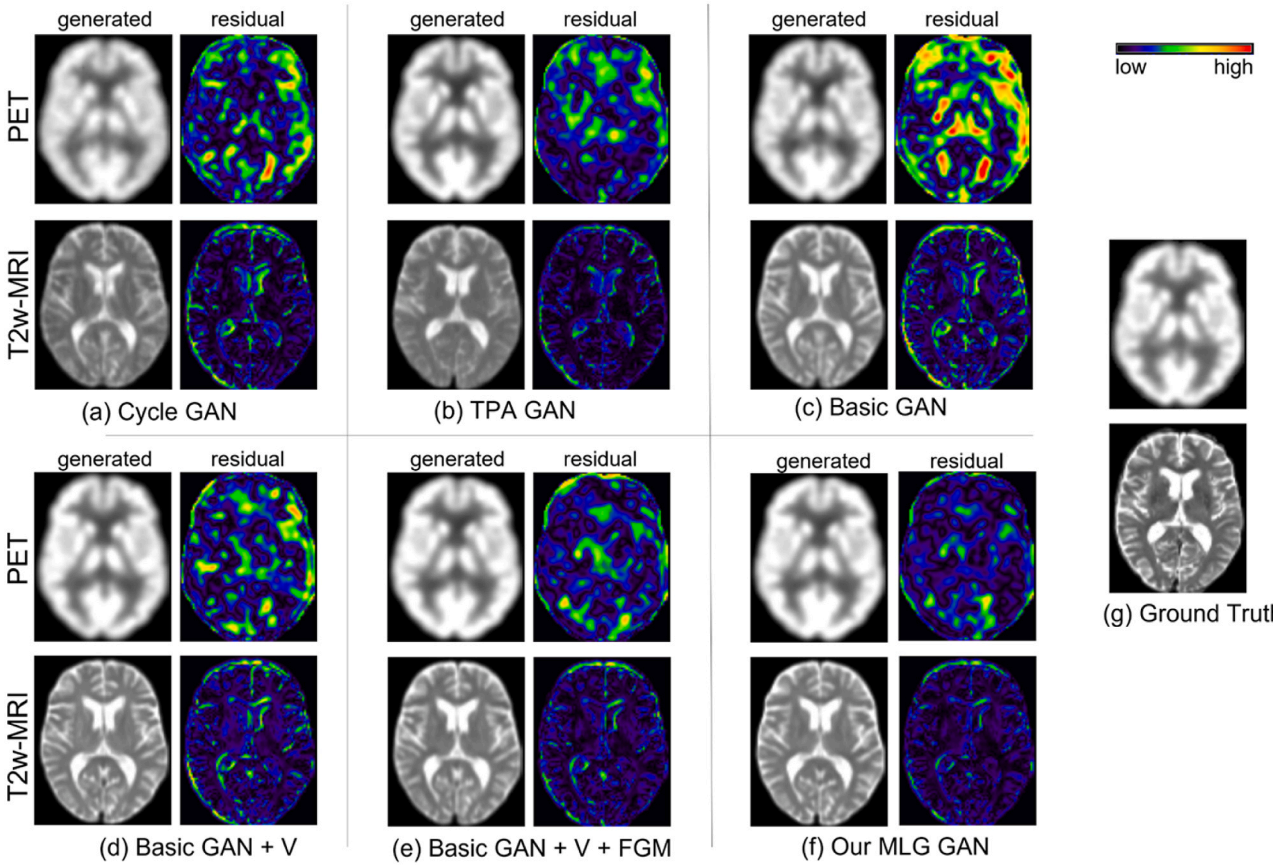


**Fig. 5.** Illustration of the generated PET and T2w-MRI for two subjects from the testing sets (i.e., ADNI-2 and OASIS-3): (a) Cycle-GAN, (b) TPA-GAN, (c) Basic GAN, (d) Basic GAN + V, (e) Basic GAN +V +FE, (f) Our MLG-GAN, and (g) The ground truth.

**Table 4**
Comparison with the single and multi-model methods for disease classification, where the models are tested on ADNI-2 and trained on ADNI-1.

| Modality | Method | AD vs CN (%) | | | | pMCI vs sMCI (%) | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | ACC | SEN | SPE | AUC | ACC | SEN | SPE | AUC |
| T1w-MRI | Single-T | 87.6 | 82.7 | 91.5 | 92.1 | 64.0 | 62.1 | 65.2 | 66.7 |
| T2w-MRI | Single-T | 84.3 | 77.8 | 91.0 | 84.5 | - | - | - | - |
| PET | Single-T | 89.4 | 88.5 | 90.0 | 94.2 | 69.1 | 69.7 | 68.8 | 62.6 |
| Multi-Modality | Aggreg | 91.3 | 89.7 | 92.5 | 95.1 | 74.1 | 72.3 | 75.3 | 78.1 |
| | Mul-T(w/o LB) | 93.0 | 91.1 | 94.5 | 96.7 | 76.5 | 73.9 | 78.5 | 81.7 |
| | Mul-T | **94.4** | **93.0** | **95.5** | **97.6** | **77.8** | **75.4** | **79.6** | **82.8** |

competitive result. Nevertheless, our MLG-GAN achieves the best performance.

### 4.3. Results of disease classification

The second experiment is to evaluate the effectiveness of multimodal classification by our Mul-T. The classification performance is quantitatively evaluated by accuracy (***ACC***), sensitivity (***SEN***), specificity (***SPE***), and area under the receiver operating characteristic (***AUC***).

First, we conduct a single-modal transformer (Single-T) network for disease classification on T1w-MRI, PET, and T2w-MRI, individually. The architecture of Single-T is the same as the task-level discriminator. Then, we take a simple aggregation (denoted as "Aggreg") at the fully connected layers of these three networks for comparison. To evaluate the effectiveness of local branches, we also conduct a Mul-T without the branches of local feature extraction (denoted as "Mul-T(w/o LB)"). The results of ablation studies are shown in Table 4, and their ROC curves are shown in Fig. 6. From the results, we can see that the Single-T on PET images has better performance than those on T1w-MRI and T2w-MRI, which suggests that PET images contain more disease-related information. The multimodal fusion methods perform better than single-modal methods. Our Mul-T benefitting from the cross-modal attention mechanism can achieve the best results. The local branches can extract more useful low-level information to further improve the classification performance. In addition, we perform the disease classification experiments on the OASIS dataset shown in Table 5. All subjects on OASIS dataset have no PET modalities, we thus generate them with our MLG-GAN. It achieves the best performance than the other methods on the OASIS dataset, although the performance seems slightly lower than that on the ADNI dataset.

Second, we compare our Mul-T with other multimodal classification methods which are both tested on ADNI-2 and trained on ADNI-1 from the baseline visits. Specifically, the comparison methods contain the nonlinear graph fusion network (Tong et al., 2017), the multimodal cascaded convolution network (Liu et al., 2018a), the landmark-based deep learning method (Liu et al., 2018c), the latent representation learning method (Zhou et al., 2019a), the spatially-constrained Fisher representation network (Pan et al., 2020) and the path-wise transfer

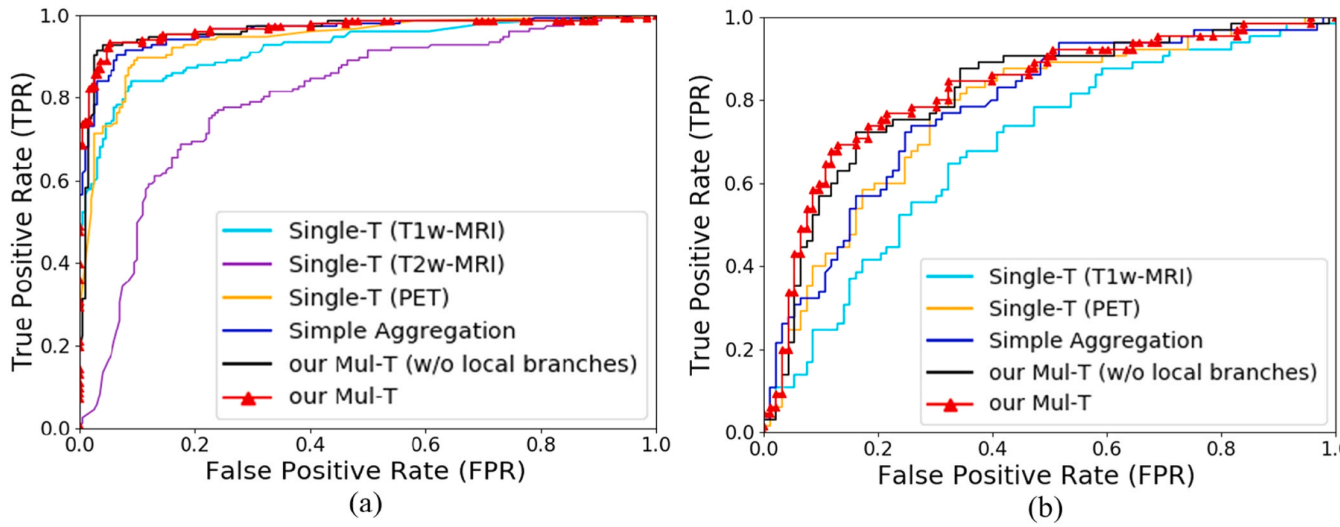


**Fig. 6.** ROC curves with different methods on T1w-MRI, T2w-MRI, PET, and Multi- modalities for disease classification: (a) AD vs CN; (b) pMCI vs sMCI. The models are tested on ADNI-2 and trained on ADNI-1.

**Table 5**
5-fold cross-validation comparison with the single and multi-model methods for disease classification on the OASIS dataset. There are no MCI patients on the OASIS dataset.

| Modality | Method | AD vs CN (%) | | | |
|---|---|---|---|---|---|
| | | ACC | SEN | SPE | AUC |
| T1w-MRI | Single-T | 84.3 | 82.5 | 86.2 | 91.0 |
| T2w-MRI | Single-T | 83.9 | 79.4 | 88.6 | 91.3 |
| PET | Single-T | 80.3 | 73.0 | 86.9 | 87.8 |
| Multi-Modality | Aggreg | 84.6 | 80.8 | 88.5 | 91.8 |
| | Mul-T (w/o LB) | 86.5 | 80.8 | **92.3** | 91.6 |
| | Mul-T | **88.4** | **84.6** | **92.3** | **94.4** |

**Table 6**
Comparison with state-of-the-art multimodal classification methods. The models are both tested on ADNI-2 and trained on ADNI-1.

| Method | AD vs CN (%) | | | | pMCI vs sMCI (%) | | | |
|---|---|---|---|---|---|---|---|---|
| | ACC | SEN | SPE | AUC | ACC | SEN | SPE | AUC |
| (Tong et al., 2017) | 88.6 | - | - | 94.8 | - | - | - | - |
| (Liu et al., 2018a) | 93.3 | 92.6 | 94.0 | 95.7 | - | - | - | - |
| (Liu et al., 2018c) | 90.9 | 87.9 | 93.3 | 96.1 | 73.5 | 74.4 | 73.4 | 80.9 |
| (Zhou et al., 2019a) | - | - | - | - | 74.3 | - | 75.5 | - |
| (Pan et al., 2020) | 93.6 | 91.5 | 95.2 | 97.0 | 77.4 | **79.1** | 77.2 | 82.5 |
| (Gao et al., 2022) | 92.0 | 89.1 | 94.0 | 95.6 | 75.3 | 77.3 | 74.1 | 78.6 |
| Our Mul-T | **94.4** | **93.0** | **95.5** | **97.6** | **77.8** | 75.4 | **79.6** | **82.8** |

dense convolution network (Gao et al., 2022). Since their works did not release the codes and relevant pre-trained models and reproducing is time- consuming, we directly use the results reported in their papers for comparison as shown in Table 6. The results show that our Mul-T achieves the best performance.

## 5. Discussion

In this section, we discuss the main differences between our proposed method and previous studies. Then, we provide the limitations of our work as well as the possible solutions in the future.

To address the problem of incomplete multimodal brain images, we construct a MLG-GAN. Compared with the previous methods (Pan et al., 2019; Gao et al., 2022; Pan et al., 2018, 2020), our proposed MLG-GAN can make full use of multi-level information for image generation. First, a task-level discriminator with transformer blocks is combined to generate the discriminative information. In addition, a feature auto-regression branch is proposed to embed the features of target images for a more accurate generation of missing data, which is different from the previous feature-consistent constraint method (Pan et al., 2019). Finally, additional voxel-wise supervision of salient regions is used to guide the generation of voxel values. Table 2 and Table 3 demonstrate their contributions, which indicates that our MLG-GAN can not only generate the details of target images but also preserve the discriminative information for disease diagnosis.

Then, a Mul-T is proposed for the final disease classification with complete multimodal brain images. In Table 6, we compare several state-of-the-art results reported in the literature for AD classification and/or MCI prognosis. For the nonlinear graph fusion method (Tong et al., 2017), the pairwise similarity was first calculated for each modality, then the similarities from multiple modalities were combined into a nonlinear graph network for the final disease classification. Liu et al. (2018a) and Liu et al. (2018c) presented the patch-based deep learning methods. Zhou et al. (2019a) raised a latent representation learning method: the complete multi-modalities were projected to a common latent representation, while the incomplete multi-modalities were projected to a modality-specific latent representation. Then these two latent representations were projected to the label space for MCI classification. Pan et al. (2020) devised a spatially constrained Fisher representation method. The innovation is that Fisher unit was conducted to represent the high-level features. Our previous work designed a hierarchical path-wise transfer-based deep learning method (Gao et al., 2022). The multi-level path-wise transfer blocks were used to make interactions between the different modalities for disease classification. Different from the above methods, we proposed a cross-modal transformer-based Mul-T, from which one modality can receive information from another modality. These alternate two-variable interactions can make full use of the complementary information between different modalities. Besides that, considering the global feature extraction on the whole brain may lose some important information related to disease, an additional branch of local feature extraction is added to aid the diagnosis of AD and MCI status.

In addition to demonstrate the performance of disease classification,

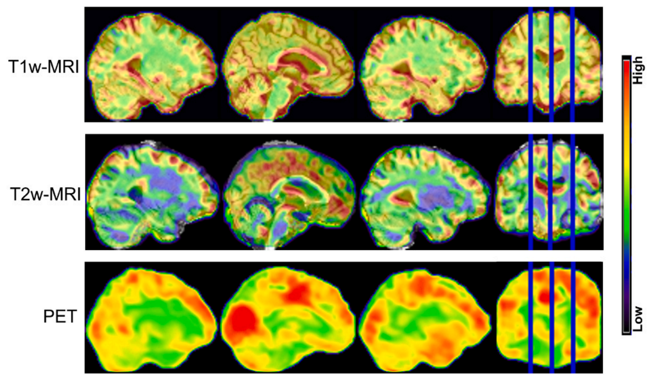


**Fig. 7.** Spatial attention maps across T1w-MRI, T2w-MRI, and PET modalities in AD diagnosis.

the brain regions with high spatial attentions learned by the model are important for interpretation because the spatial attention indicates the weights of local regions. For better visualization, we overlay the spatial attention maps to a brain template as shown in Fig. 7. We can see that the regions with high attentions in T1w-MRI image mainly appear in Temporal Lobe, Corpus Callosum, Lateral Ventricle, Parahippocampal Gyrus and Parietal Gyrus; in terms of T2w-MRI image, the highlight regions mainly appear in Lateral Ventricle, Cingulate Gyrus and Frontal Gyrus; while to the PET image, the highlight regions mainly appear in Cingulate Gyrus, Cuneus, and Occipital Lobe. Above findings suggest that multimodal images could provide powerful and complementary information for more accurate disease diagnosis. The findings from the medical literature (Clerx et al., 2013; Shimoda et al., 2015; Etminani et al., 2022) show highly consistent with our highlight brain regions, which might be useful as reference for clinical professionals.

However, there are some limitations in our work that may be addressed in the future. First, except for the T1w-MRI, PET, and T2w-MRI, other clinical data such as CSF and SNP could be contained to further improve the performance of disease classification. Second, the data acquisition protocols in the ADNI database are significantly different from those in OASIS. Our proposed method did not consider the differences, and we used the same pipeline to process the images of these two databases. In future works, data adaption techniques would be studied to address this problem.

## 6. Conclusion

In this paper, we proposed a multi-level guided GAN (MLG-GAN) and a multimodal transformer (Mul-T). The MLG-GAN combines the task-level discriminator, feature-level embedding, and voxel-level supervision for a more accurate generation of missing images. With the complete multimodal images, Mul-T uses the cross-modal transformers to model the interactions of multi-modal modalities and then combine the high-level features from global and local branches for disease diagnosis. Comprehensive experiments on three independent datasets demonstrated the effectiveness of our method.

## CRediT authorship contribution statement

**Xingyu Gao:** Conceptualization, Methodology, Software, Validation, Visualization, Data Curation, Writing-Original draft preparation. **Feng Shi**: Supervision, Writing-Reviewing and Editing. **Dinggang Shen**: Supervision, Writing-Reviewing and Editing. **Manhua Liu:** Conceptualization, Methodology, Formal analysis, Writing-Reviewing and Editing, Project administration, Funding acquisition.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

The data that has been used is confidential.

## Acknowledgments

This work was supported in part by the National Natural Science Foundation of China under Grant 62171283, in part by the Natural Science Foundation of Shanghai under Grant 20ZR1426300, in part by the National Key Research and Development program of China under Grant 2022YFC2503302, 2022YFC2503305 and 2022YFE0205700, in part by the CAAI-Huawei MindSpore Open Fund, Shanghai Municipal Science and Technology Major Project under Grant 2021SHZDZX0102 and 2018SHZDZX01, and in part by the Fundamental Research Funds for the Central Universities. and the Key Laboratory of Computational Neuroscience and Brain-Inspired Intelligence (LCNBI) and ZJLab.

We also thank the ADNI and OASIS very much for their data collection and sharing. For more information, we refer readers to website of the ADNI: https://adni.loni.usc.edu/ and OASIS:https://www.oasis-brains.org/.

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